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Iterated Rationality Models with Lexical Uncertainty

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1 Abstract

Recent works present models of game-theoretic approaches to derive Scalar Implicatures (SI), within the neo-Gricean approach, which can be referred as Iterated Rationality Models (IRM). In particular, the Rational Speech Act (RSA) framework assumes a probabilistic inference game between a pragmatic speaker and listener. One of the successful versions of RSA are of Bergen, Levy, and Goodman (2016), assuming compositional *lexical uncertainty*. However, a lexicalist account for SI is prone to internal conflicts, such as prediction of suspension under negation, divergence in DE contexts, wrong predictions for two-upper-bound triggering expressions and for ellipsis. A naive lexical ambiguity approach does not seem to escape these problems. In this work, I review the lexical uncertainty approach for neo-Gricean pragmatics within the IRM framework. For simplicity, I assume the IRM framework of Fox and Katzir (2021). I define a lexical uncertainty implementation of Fox and Katzir (2021), and test the predictions of this model for various implicatures that were suggested to challenge the lexicalist account.

2 Introduction

Scalar implicatures are implicit inferences of explicit utterances. Consider, for example, the simple inference in (1):

(1) John met some student.

$\rightsquigarrow$ John met some but not all of the students.

Here, 'some' is interpreted as 'some but not all'. Any theory of SI should be able to explain such upper bound inferences. The simplest theory one can suggest is semantic lexical encoding. In this theory, the lexical upper bound implicature is conjuncted with the original semantic value of 'some'. Thus, we can define

(2) $[\![\text{some student}]\!](P) = 1$ iff $\exists x[\text{student}(x) \wedge P(x)] \wedge \neg\forall x[\text{student}(x) \rightarrow P(x)]$

Now, the lexical value of 'some' is strengthened to contain the upper bound, and we can use the new semantic value to derive the implicature in (1). However, encoding implicatures in the lexicon could lead to undesired results. For instance, note that upper bound implicatures can be suspended, as in (3)

(3) John met some student. In fact, John met all the students.

If we assume that the upper bound inference is part of the lexical value of *some*, we get that *John met some but not all of the students* and *John met all of the students*, which is a contradiction. Note that unlike implicatures, The assertion cannot be suspended:

(4) # John met some student. In fact, John didn't meet any student.

Another problem of the lexicalist approach is the interpretations of downward entailing (DE) contexts, and negation in particular. Let's look at the negation of (1) (taking 'some' in narrow scope):

(5) John did not meet a/some student.

If we take the lexical value of 'some' to be (2), then the meaning of (5) would be that John did not meet any student (the negation of the first half of (2)), or that John met all students (the negation of the second half of (2)). This divergence under negation is not the natural interpretation of (5), where the upper bound is not expected.

A possible suggestion to overcome these challenges to the lexicalist approach, is assuming lexical ambiguity between the literal semantic value and the upper bound interpretation. One lexical value could be used for upward entailing (UE) contexts, and the other could be used for DE contexts. In our example, we can split the lexical entry of 'some':

(6) a. $[\![\text{some}_1\ \text{student}]\!](P) = 1$ iff $\exists x[\text{student}(x) \wedge P(x)]$

b. $[\![\text{some}_2\ \text{student}]\!](P) = 1$ iff $\exists x[\text{student}(x) \wedge P(x)] \wedge \neg\forall x[\text{student}(x) \rightarrow P(x)]$

Now, we can use $[\![\text{some}_2]\!]$ in UE contexts, except for suspensions where we fallback to $[\![\text{some}_1]\!]$, and we use $[\![\text{some}_1]\!]$ in DE contexts to avoid undesired divergence. This could solve both problems. An elegant rule to choose between these two lexical definitions, is simply to always choose the stronger meaning, which is $[\![\text{some}_1]\!]$ in DE contexts and $[\![\text{some}_2]\!]$ in UE contexts.

Nevertheless, lexical ambiguity still makes wrong predictions. First, If one assumes that the rule for choosing between the two lexical values is always to take the stronger meaning, one could say that suspension, like in (3), must involve reanalysis from $[\![\text{some}_2]\!]$ to $[\![\text{some}_1]\!]$. However, the opposite reanalysis is not possible. We can see it under negation, where suspension is not allowed:

(7) # John did not meet a/some student. In fact, John met all the students.

Again we see a core asymmetry between the assertion and the upper bound inference.

Moreover, there are implicatures that do not seem to have any lexical source, at least given our definitions so far. For example, implicatures in DE contexts (where 'every' takes narrow scope):

(8) John didn't meet every student.

$\rightsquigarrow$ John met at least one student.

In later sections, I elaborate about other problems of the lexical ambiguity approach, such as expressions that trigger more than one upper bound, or the semantic identity in ellipsis. Those challenges to the lexical approach are an argument in favor of modern approaches to SI that derive implicatures by negating alternatives, such as the neo-Gricean theory (Sauerland, 2004) or the grammatical theory (Chierchia, Fox, and Spector, 2012).

Both the neo-Gricean approach and the grammatical approach derive SI by conjuncting the base assessment with some negation of alternatives. While the neo-Gricean approach assumes pragmatic reasoning and global negation of alternative utterances, the grammatical approach assumes composition and local exhaustification.

Within the neo-Gricean approach, recent suggestions implement pragmatic Gricean reasoning with game-theoretic modeling, combining a rational speaker and listener that iteratively adjust their assumptions about the reasoning of the other. Following Fox and Katzir (2021) I refer to this collection of methods as Iterated Rationality Models (IRM). Such suggestions include the Iterated Best Response (IBR) model of Franke (2011), and the Rational Speech Act (RSA) framework of Frank and Goodman (2012).

A common extension of the RSA framework was suggested by Bergen, Levy, and Goodman (2016). Bergen, Levy, and Goodman (2016) propose a realignment between semantics and pragmatics. In their RSA variant, the semantic value of an utterance is not fixed. Rather, similarly to the lexical ambiguity theory, each utterance has multiple lexical values which are combined to create the literal listener's initial interpretations. Bergen, Levy, and Goodman (2016) show that this framework, named RSA with lexical uncertainty, derives both M-implicatures and embedded implicatures which violate Hurford's constraint, both cannot be derived by the original RSA framework. In a following work, Potts et al. (2016) present RSA with lexical uncertainty as a pragmatic framework to derive embedded implicatures, which could serve as an argument in favor of pragmatic theories for SI over the grammatical approach. In addition, Champollion, Alsop, and Grosu (2019) extended Bergen, Levy, and Goodman (2016) to derive also free choice disjunction inferences.

While RSA is a pragmatic framework, the use of multiple lexica to define semantic value in Bergen, Levy, and Goodman (2016) is similar to the lexical ambiguity theory we have just presented. Thus, this popular theory might face the same challenges of the lexicalist account to SI. This is what I aim to examine in this work. I construct a simplified version of Bergen, Levy, and Goodman (2016), based on a different pragmatic framework suggested by Fox and Katzir (2021). Then, I test the power of this theory by deriving inferences that are known to challenge lexical ambiguity.

This work is constructed as follows: In section 3, I provide the necessary background to understand the modeling of Fox and Katzir (2021). In section 4, I present my suggested model, adding the lexical uncertainty component over Fox and Katzir (2021). I derive some basic predictions and discuss some extensions and setting choices. In section 5, I show how does my suggested model deal with the challenges of lexicality that were presented above. Lastly, I discuss some arguments against lexical uncertainty.

3 Iterated Rationality Models

The pragmatic IRM framework I use in this work is not RSA, but a simplified version of Franke (2011) suggested by Fox and Katzir (2021). The use of a simpler model, compared to the RSA framework, allows us to focus on the model assumptions and to see whether the predictions of the model are robust to minor changes in the assumptions.

The assumptions of the IRM framework in Fox and Katzir (2021) can be

described as follows:

(9) a. The possible worlds in the pragmatic context are partitioned into cells, which represent the speaker epistemic states.
b. The epistemic state of the speaker entails one of the states (opinionatedness assumption).
c. The speaker's goal is to convey the listener to their epistemic state.
d. The speaker is truthful, adhering to Grice's Maxim of Quality.

Let's notate the set of possible epistemic states with Π. The states are induced by the speaker's set of alternative messages, which we notate with M. If not said otherwise, I will assume the structural suggestion of Katzir (2007) to derive alternatives, however note that Bergen, Levy, and Goodman (2016) do not apply this suggestion in their formulation.

In the IRM framework, the pragmatic reasoning of the speaker and the listener is done by iteratively pairing messages and epistemic states. Let's start with the simplest example, the strengthening of 'some' to 'some but not all', as in (1). In this case, the messages contain the alternatives "some" and "all". Thus, the induced speaker states are $\neg\exists$, $\exists$ and $\exists \wedge \neg\forall$. Note that generally we can ignore states that could never be conveyed with any message, like $\neg\exists$ in this case. I do not include them in the following examples.

Now, the speaker can convey their state as follows: In the first step, If the speaker epistemic state is $\forall$, they can say "all". When the listener hears the message "all", only the state $\forall$ is consistent with this message, so the listener can select the correct cell. In the second step, If the speaker's epistemic state is $\exists \wedge \neg\forall$, they know that they can rely on the listener knowledge in the previous step. That means, the listener knows that if the speaker epistemic state was $\forall$, the speaker would have used the message "all". Therefore in this case the speaker would use the message "some". The listener, who can rule out the cell $\forall$, is left with the cell $\exists \wedge \neg\forall$.

A general way to apply this strategy is identifying an epistemic state with a specific message. Formally, we use this definition of state identification:

(10) State Identification (non-probabilistic version): A message $m \in M$ identifies a state $t \in \Pi$ if m is true in t, and there is no other $t' \in \Pi$ such that m is true in t'.

Now, the pragmatic strategy is applied iteratively, paring messages and states. The listener starts with the set of all messages and states. At each iteration, states are identified by messages according to the state identification criterion. After the identification, those states and messages are removed, and the next iteration starts with the remaining states and messages. When no more state identifications are possible, the algorithm stops.

In our example, in the first iteration the state $\forall$ is identified with the message "all", and then this message and state are removed. In the second iteration, only the state $\exists \wedge \neg\forall$ and the message "some" are left, and identification is

possible in this case. Without the elimination of "all", the state $\exists \wedge \neg\forall$ would not be identified.

This strategy of state identification can be summarized in a table formulation, where the columns correspond to the set of alternative messages M and the rows correspond to the set of epistemic states Π. Each cell in the table indicates whether the relevant message is consistent with the relevant state of not, using 1 and 0 respectively. It is common to ignore states that will result 0 for all messages, like $\neg\exists$ here. The colors of the cells indicate the iteration where the cell has been identified or eliminated. In our strengthening 'some' to 'some but not all' example, the resulted table is this:

	'some'	'all'
$\exists \wedge \neg\forall$	1	0
$\forall$	1	1

Table 1: 'some'/'all' with non-probabilistic IRM

identified message-state pairs in iteration I
cells eliminated after iteration I
identified message-state pairs in iteration II

3.1 Probabilistic IRM

The simplified version of IRM presented above is a formal strategy for the speaker to convey a listener with their epistemic state. While this version of IRM is able to derive the basic implicature in (1), it does not predict more complicated implicatures. An example of this is the free choice disjunction in (11).

(11) John is allowed to eat an apple or a banana.

$\rightsquigarrow$ John is allowed to eat an apple.

$\rightsquigarrow$ John is allowed to eat a banana.

In regular disjunctions, an individual disjunct is not entailed by the disjunction. However, in free choice disjunction, each disjunct is an inference of the disjunction, and claimed to be an implicature (Kamp, 1974). More broadly, This free choice phenomenon is more general and is called Conjunctive Reading of Disjunction (CRD). As Fox and Katzir (2021) show, the basic version of IRM does not derive the CRD of 2 elements. They use it as an argument in favor of the probabilistic version of IRM.

In the probabilistic version, the listener does not assume certainty about the speaker intentions, but rather the listener uses the best guess they can infer about the speaker's epistemic state. Intuitively, the listener would guess the state with the highest probability.

To implement a probabilistic version of IRM, the state identification criterion should be updated. We can define the probability measure $p(m|t)$ which is the probability that the speaker will utterance a message m given that their epistemic state is t. In our 'some'/'all' case, in the support of p we have $p(\text{all}|\forall) > 0$ for "all", and both $p(\text{some}|\exists \wedge \neg\forall) > 0$, $p(\text{some}|\forall) > 0$ for "some".

A common assumption in the IRM literature, and also in Fox and Katzir (2021), is that the speaker has no preferences between possible true messages given a certain state. Hence, in this case, both $p(\text{some}|\forall) = p(\text{all}|\forall) = \frac{1}{2}$ and $p(\text{some}|\exists \wedge \neg\forall) = 1$. Generally, this assumption is referred as a *naive speaker*.

(12) Naive Speaker: if cell t makes n different messages true, the speaker will choose each of them with equal probability $\frac{1}{n}$

Note that in the IRM literature, probabilities are often used to predict the representations of the speaker and listener's world knowledge. Thus, some theories aim to have high correlation between the model probabilities and human interpretations, such as Potts et al. (2016). In this work however I treat probabilities as a formal tool to derive implicatures, and not as an approximation for human epistemic states. See Fox and Katzir (2021) for further discussion.

In any case, the new state identification criterion in the probabilistic version of IRM should be refined to include probabilities. When the listener receives a message m from the speaker, they have to compare all possible states that could lead to this message, thus computing the posterior $p(t|m)$ for all states. According to Bayes theorem, $p(t|m)p(m) = p(m|t)p(t)$. To compute the likelihood $p(m|t)$, we rely on another common assumption of *flat priors*:

(13) Flat Priors (for states): The state prior distribution is uniform. Thus if Π is finite then $p(t) = \frac{1}{|\Pi|}$ for any $t \in \Pi$.

All is left is defining the probabilistic state identification criterion. Intuitively, we would say that message m identifies state t if the posterior is maximal, meaning for every $t' \in \Pi$ we have $p(t'|m) < p(t|m)$. Now that we assume flat priors, this is the same as comparing the likelihoods:

(14) State Identification (probabilistic version): A message $m \in M$ identifies a state $t \in \Pi$ if for every other $t' \in \Pi$, $p(m|t) > p(m|t')$.

Back to the example, assuming flat priors we can compare the likelihoods. Using the new state identification criterion, we get that "all" identifies $\forall$, since $p(\text{all}|\forall) > p(\text{all}|\exists \wedge \neg\forall)$ and similarly "some" identifies $\exists \wedge \neg\forall$. We can use a probabilistic version of the table format for summarization, and see that all states are now identified in a single iteration.

$p(m\|t)$	'some'	'all'
$\exists \wedge \neg\forall$	1	0
$\forall$	$\frac{1}{2}$	$\frac{1}{2}$

Table 2: 'some'/'all' with probabilistic IRM

identified message-state pairs in iteration I
cells eliminated after iteration I

Fox and Katzir (2021) show that this probabilistic version of IRM does derive CRD with two elements, in contrast to the non-probabilistic version.

4 IRM with Lexical Uncertainty

Equipped with the necessary background on IRM, I can now present the main suggested model to examine in this work. The main claim of Bergen, Levy, and Goodman (2016) I try to extend in this work, is that semantic content might be underspecified, and pragmatic reasoning can affect the interpretation of the semantic content. Thus, the implementation of Bergen, Levy, and Goodman (2016) involves merging of the semantic content of different lexica into the pragmatic reasoning.

So far, we assumed a fixed lexicon shared to the speaker and listener and one semantic value for each word, e.g. "some" or "all". Following Bergen, Levy, and Goodman (2016), I will now define a simplified version of *lexical uncertainty*, with a model I call IRM with lexical uncertainty (IRM-LU). In this version, we have a set of lexica Λ, with a prior probability $p(L)$ to select each lexicon $L \in \Lambda$.

In the speaker-listener interaction, the listener does not know which lexicon the speaker uses. Thus, instead of the regular likelihood $p(m|t)$, the likelihood is now conditioned on the lexicon L the speaker uses, $p(m|t, L)$. Yet, this is not a major change to the probabilistic IRM framework: the overall likelihood $p(m|t)$ can be reconstructed by marginalization over all possible lexica. Thus, we get:

$$p(m|t) = \sum_{L \in \Lambda} p(m|t, L)p(L|t)$$

Here, $p(L|t)$ is the probability that the speaker will choose a lexicon L given that their epistemic state is $t \in \Pi$. I assume that if L does not contain any message $m \in M$ that satisfies t, the speaker will not use that lexicon. Over the remaining lexica, I assume a uniform distribution. Overall, we can summarize this is the *flat likelihoods* assumption:

(15) Flat Likelihoods (for lexica): For any state $t \in \Pi$, If n lexica contain a message $m \in M$ that satisfies t, the likelihood $p(L|t)$ of choosing each lexicon is $\frac{1}{n}$. If L does not contain a message to satisfy t, then $p(L|t) = 0$.

After computing $p(m|t)$ by lexicon marginalization for every message and state, the iterative probabilistic state identification continues the same. Note

that on the speaker side, the state t is also conditioned on a selected lexicon, and a similar marginalization can be applied to compute $p(t|m)$, however it is not required under the previous naive speaker and flat prior assumptions.

While the general assumption of lexical uncertainty allows for arbitrary lexica, following Bergen, Levy, and Goodman (2016), I will also use the largest set of lexica Λ that is possible with the base semantics. Let L_s be a base lexicon which contains the literal semantic values of words such as "some" and "all". We would like that a proper new lexicon to enrich Λ will be any lexicon that entails L_s, and it is non-contradictory. We call such lexicon a *refinement* of the base lexicon L_s.

For simplicity, in this section I assume that every possible message $m \in M$ has a semantic value in every lexicon $L \in \Lambda$. Later, I define how to compose semantic values from atomic entries. I notate $L(m,t) = 1$ if state $t \in \Pi$ is true under message $m \in M$ and lexicon $L \in \Lambda$.

Finally, I use this definition for lexical refinement:

(16) Lexical Refinement: A lexicon L' refines lexicon $L \in \Lambda$ if for every state $t \in \Pi$ and message $m \in M$, if $L'(m,t)$ then $L(m,t) = 1$. Also, there is a state $t \in \Pi$ and a message $m \in M$ such that $L'(m,t) = 1$.

Starting with the literal lexicon L_s, we can create $\tilde{\Lambda}$ which contains all possible refinements of L_s. This will be the default set of lexica for pragmatic computation, and will be assumed in the following derivations.

4.1 Simple Predictions

To illustrate the basic definitions of IRM-LU, I demonstrate the derivations of strengthening "some" to "some but not all" as in (1). Let's start with the default lexicon $L_s = L_1$ which contains the default semantic values:

$$L_1 = \left\{ \begin{array}{c} [\![\text{all}]\!] = \{\forall\} \\ [\![\text{some}]\!] = \{\exists \wedge \neg\forall, \forall\} \end{array} \right\}$$

When we search for possible lexical refinements, there are two additional refined lexica to add to Λ:

$$L_2 = \left\{ \begin{array}{c} [\![\text{all}]\!] = \{\forall\} \\ [\![\text{some}]\!] = \{\exists \wedge \neg\forall\} \end{array} \right\} \quad L_3 = \left\{ \begin{array}{c} [\![\text{all}]\!] = \{\forall\} \\ [\![\text{some}]\!] = \{\forall\} \end{array} \right\}$$

For each lexicon, we can compute the likelihoods $p(m|t,L)$. Using the probabilistic table format, we get:

$p(m\|t, L_1)$	'some'	'all'
$\exists \wedge \neg\forall$	1	0
$\forall$	$\frac{1}{2}$	$\frac{1}{2}$

$p(m\|t, L_2)$	'some'	'all'
$\exists \wedge \neg\forall$	1	0
$\forall$	0	1

$p(m\|t, L_3)$	'some'	'all'
$\exists \wedge \neg\forall$	0	0
$\forall$	$\frac{1}{2}$	$\frac{1}{2}$

Now we can marginalize over all lexica. If the epistemic state of the speaker is $\forall$, the uncertainty is applied over all possible lexica in Λ, and by the flat likelihood assumption, $p(L|\forall) = \frac{1}{3}$. However, If the epistemic state of the speaker is $\exists \wedge \neg\forall$, the listener knows that L_3 is not a valid lexicon for the speaker, therefore L_3 will not take part in the computation, since by our assumption $p(L_3|\exists \wedge \neg\forall) = 0$. Thus, we get the overall likelihoods:

$$
\begin{aligned}
p(\text{some}|\exists \wedge \neg\forall) &= \frac{1}{2} * 1 + \frac{1}{2} * 1 + 0 * 0 = 1 \\
p(\text{some}|\forall) &= \frac{1}{3}(\frac{1}{2} + 0 + \frac{1}{2}) = \frac{1}{3} \\
p(\text{all}|\exists \wedge \neg\forall) &= \frac{1}{2} * 0 + \frac{1}{2} * 0 + 0 * 0 = 0 \\
p(\text{all}|\forall) &= \frac{1}{3}(\frac{1}{2} + 1 + \frac{1}{2}) = \frac{2}{3}
\end{aligned}
$$

And the marginalized likelihood table $p(m|t)$ is:

$p(m\|t)$	'some'	'all'
$\exists \wedge \neg\forall$	1	0
$\forall$	$\frac{1}{3}$	$\frac{2}{3}$

Table 3: 'some'/'all', IRM with lexical uncertainty (IRM-LU)

identified message-state pairs in iteration I
cells eliminated after iteration I

Similarly to the basic probabilistic version of IRM, we obtain the correct state identification in one iteration.

4.2 Compositionality

An important component of the IRM-LU model is handling compositional messages. So far, we assumed that each speaker message $m \in M$ has a lexical entry in each lexicon $L \in \Lambda$, and in practice we build Λ from refinements of the initial literal lexicon L_s. Of course, this assumption is not realistic as it requires

numerous lexical entries, and does not let the compositional semantics to be applied. A more reasonable view is to apply lexical refinements over atomic lexical elements, and build the speaker messages meanings with compositionality rules.

Again, I will assume Bergen, Levy, and Goodman (2016) approach to compositionality which is quite straight-forward. A full compositional mechanism is not in the scope of this work, and I will focus on simple binary composition of atomic elements. We start again with the literal semantic lexicon L_s. Now we assume it to contain also mappings of atomic messages M_a to boolean truth values. Similarly, we can create the set of lexica Λ from L_s using lexical refinements of atomic lexical values. The denotations in the lexicon are defined inductively. For atomic messages $m \in M_a$, the denotation $[[m]]$ in lexicon $L \in \Lambda$ is identical to its entry in the lexicon:

$$[[m]]_L(t) = L(m, t)$$

For binary operators, such as "or", we use the denotation of the binary operator in the lexicon, and can similarly define the denotation

$$[[m_1 \text{ or } m_2]]_L(t) = [[\text{or}]]_L\left([[m_1]]_L(t), [[m_2]]_L(t)\right)$$

The binary operators can also be refined. This can be similarly extended to more compositional denotations, and sentence meanings will be derived with known semantic compositional rules.

4.3 Conjunctive Reading of Disjunctions

Recall the free choice inference from (11). As mentioned, in free choice disjunction, both disjuncts are inferences of the disjunction, and this is an example for conjunctive reading of disjunction (CRD).

(11) John is allowed to eat an apple or a banana.

$\rightsquigarrow$ John is allowed to eat an apple.

$\rightsquigarrow$ John is allowed to eat a banana.

The probabilistic IRM suggested by Fox and Katzir (2021) was shown to derive CRD with two disjuncts. Now I show that also IRM-LU can derive CRD with two disjuncts. Using the notation *John is allowed to eat an apple* by message A and *John is allowed to eat a banana* by message B, the message in (11) is 'A or B'. In CRD, the alternative messages set is not closed to conjunction, hence we assume the alternatives are {'A', 'B', 'A or B'}. The induced states for CRD are $\{A \wedge \neg B, B \wedge \neg A, A \wedge B\}$. To add the lexical uncertainty component, we note that while 'and' has no further lexical refinement, there are three possible lexical refinements of 'or': the literal lexicon where [[or]] is a simple disjunction, the lexicon where [[or]] is interpreted as exclusive, and the lexicon where [[or]] is refined to 'and'.

Skipping some intermediate computations, the marginalized likelihood is obtain in Table 4. We see that in the first iteration, 'A' identifies the state

$p(m\|t)$	'A'	'B'	'A or B'
$A \wedge \neg B$	$\frac{2}{3}$	0	$\frac{1}{3}$
$B \wedge \neg A$	0	$\frac{2}{3}$	$\frac{1}{3}$
$A \wedge B$	$\frac{7}{18}$	$\frac{7}{18}$	$\frac{2}{9}$

Table 4: CRD with two disjuncts, IRM-LU

cells eliminated after iteration I
identified message-state pairs in iteration I
identified message-state pairs in iteration II

$A \wedge \neg B$ and 'B' identifies the state $B \wedge \neg A$. Thus, after the elimination, in the second iteration 'A or B' identifies the remaining state $A \wedge B$, providing the CRD reading.

Fox and Katzir (2021) showed that while it seems that IRM can derive CRD, the model fails to derive CRD for a greater number of disjuncts. Later, Frischoff (2025) suggested an enhanced version of IRM that does generalize to more disjuncts. Here I do not assume the version of Frischoff (2025), and test the consequences of adding lexical uncertainty to IRM. It turns out that IRM-LU, similarly to the IRM version of Frischoff (2025), does derive CRD for $n \geq 3$ disjuncts. While a proof for this fact is out the scope of this work, I demonstrate the derivation of three disjuncts.

For three disjuncts, the message is 'A or B or C', and the alternatives are {'A','B','C', 'A or B', 'A or C', 'B or C', 'A or (B or C)'}, assuming the structural definition of Katzir (2007). Assuming the same set of three lexica as before, we obtain the IRM-LU probabilities in Table 5. In this case, in the first iteration the

$p(m\|t)$	'A'	'B'	'C'	'A or B'	'A or C'	'B or C'	'A or B or C'
$A \wedge \neg B \wedge \neg C$	$\frac{1}{2}$	0	0	$\frac{1}{6}$	$\frac{1}{6}$	0	$\frac{1}{6}$
$\neg A \wedge B \wedge \neg C$	0	$\frac{1}{2}$	0	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{6}$
$\neg A \wedge \neg B \wedge C$	0	0	$\frac{1}{2}$	0	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
$A \wedge B \wedge \neg C$	$\frac{1}{4}$	$\frac{1}{4}$	0	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{5}{36}$	$\frac{1}{18}$
$A \wedge \neg B \wedge C$	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{18}$
$\neg A \wedge B \wedge C$	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{5}{36}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{1}{18}$
$A \wedge B \wedge C$	$\frac{5}{28}$	$\frac{5}{28}$	$\frac{5}{28}$	$\frac{2}{21}$	$\frac{2}{21}$	$\frac{2}{21}$	$\frac{5}{28}$

Table 5: CRD with 3 disjuncts, IRM-LU

cells eliminated after iteration I
cells eliminated after iteration II
identified message-state pairs in iteration I
identified message-state pairs in iteration II

messages 'A','B','C' identify the states of only A, only B and only C respectively,

and the message 'A or B or C' identifies the conjunction $A \wedge B \wedge C$, providing the CRD derivation. The rest of the states are identified in the second iteration.

While IRM-LU derives CRD for $n \geq 3$ disjuncts in this setting, it is worth noting that the calculations are highly dependent on the set of lexica Λ. A different set of lexica (such as omitting the refinement where [[or]] is interpreted as a conjunction) might not derive this result.

4.4 Granularity of the Lexical Ambiguity

In my construction of the IRM with lexical ambiguity model, I assumed, following Bergen, Levy, and Goodman (2016), a set of lexica Λ. Each lexicon $L \in \Lambda$ is consistent, thus the interpretation of for example 'some' is consistent within each lexicon. I could have defined the lexical uncertainty differently, where I assume a single lexicon $\hat{L}$ that has lexical ambiguity for values, and the refinement is applied within the same lexicon. For example, $\hat{L}$ can contain both the literal interpretation of 'some' and the upper bound interpretation. Then, we could have used the single lexicon $\hat{L}$ with the probabilistic IRM of Fox and Katzir (2021), where different messages are generated from different lexical interpretations, involving compositionality.

Eventually I did not choose that implementation to lexical uncertainty. Here I show an argument in favor of the selected definition of IRM-LU, based on predictions for CRD with two disjuncts. Back to example (11), the message is 'A or B'. Let's assume that we have a single lexicon, with lexical ambiguity for 'or'. Hence $[[\text{or}_1]] \in \hat{L}$ is a disjunction, and $[[\text{or}_2]] \in \hat{L}$ is exclusive or. The structural alternatives in this case will be {'A', 'B', 'A or_1 B', 'A or_2 B' }, without conjunction. Now, we obtain the following probabilities:

$p(m\|t)$	'A'	'B'	'A or_1 B'	'A or_2 B'
$A \wedge \neg B$	$\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{3}$
$B \wedge \neg A$	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$A \wedge B$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0

The result is that no state is identified, since for every message m there is no maximal state t for $p(m|t)$. In particular, $A \wedge B$ is not identified. Thus, this IRM with a single refined lexicon cannot derive CRD for 2 disjuncts.

5 Dealing with Lexicality Challenges

Up to now, I have presented the setting for IRM-LU, merging ideas and definitions from Fox and Katzir (2021) and Bergen, Levy, and Goodman (2016). I demonstrated some basic derivations of this model. Now that the model is defined, we can examine how does IRM-LU handle the challenges of the lexical account for deriving implicatures that were presented in the introduction.

5.1 Implicature Suspension

Recall from (3) that implicatures can be suspended in UE contexts:

(3) John met some student. In fact, John met all the students.

This was a problem for the first lexical account, which holds a single lexical value for the literal interpretation and the upper bound interpretation of "some". Following that, we suggested the lexical ambiguity approach to include two lexical values for "some": $[[\text{some}_1]]$ for the literal meaning and $[[\text{some}_2]]$ for "some but not all". This ambiguity is similar to the lexical refinement of our model, since the semantic value $[[\text{some}]] = \{\exists \wedge \neg\forall, \forall\}$ of the literal lexicon can be refined to $[[\text{some}]] = \{\exists \wedge \neg\forall\}$, and cause a contradiction in (3).

In the lexical ambiguity account, we suggested always selecting the stronger interpretation. This explains basic UE implicatures like (1). In the case of suspension, we assume reanalysis given the second sentence in (3), retracing back to the weaker $[[\text{some}_1]]$. However, the rule of selecting the stronger interpretation could not explain why suspension under negation is not possible. Even if we assume reanalysis, the weaker interpretation in DE contexts, which is $[[\text{some}_2]]$, still provides a contradiction in (7).

(7) John did not meet a/some student. In fact, John met all the students.

It seems that in Bergen, Levy, and Goodman (2016) and in our IRM-LU model this problem does not reoccur. This is since there is not a single rule to select a lexicon. Rather, the listener incorporate all lexica in their pragmatic computation, and rules out the contradictions using $p(L|t)$. We can assume in a simplified manner the composition of the suspension in (3) as the conjunction [[some and all]]. Using the compositional model, this conjunction is a contradiction if the semantic value of "some" is $\{\exists \wedge \neg\forall\}$, and the conjunction is $\{\forall\}$ for the literal "some" which is $\{\exists \wedge \neg\forall, \forall\}$. When the listener hears the suspension in (3), they incorporate all possible lexica in Λ. They assume that the speaker would not have chosen a lexicon that makes their utterance contradictory. In other words, $p(L|t) = 0$ if L does not contains a message to satisfy t. This is the listener's pragmatic step to rule out the contradictory lexicon. Assuming the alternatives Alt('some and all') = {'some', 'all', 'some and all'} (adding the disjunction alternative does not change the results), and the lexica set $\Lambda = \{L_1, L_2, L_3\}$ from the previous 'some'/'all' example, we obtain the following table after computing the marginalization of $p(m|t)$.

$p(m\|t)$	'some'	'all'	'some and all'
$\exists \wedge \neg\forall$	1	0	0
$\forall$	$\frac{2}{9}$	$\frac{5}{9}$	$\frac{2}{9}$

Table 6: 'some'/'all' implicature suspension, IRM with lexical uncertainty

identified message-state pairs in iteration I
cells eliminated after iteration I

In the resulting table, the message "some and all" for the suspension in (3) indeed identifies $\forall$ with lexical ambiguity, together with "all" (Note that under the formulation of Fox and Katzir (2021), more that one message can identify a state in a single iteration).

In particular, IRM-LU does not predict the impossible suspension under negation. Unlike the lexical rule of taking the strongest meaning, here all meanings are combined to determine the correct state. Overall, we see that a pure lexical account is not enough. The lexical uncertainty model combines lexical uncertainty with pragmatic reasoning on the listener side.

5.2 Negation

Recall that in DE contexts, and negation in particular, presented other challenges for the lexical account for SI. The first problem was unpredicted divergence under negation when assuming the upper bound is part of the lexical value, such as in (5).

(5) John did not meet a/some student.

If we assume 'some' to be 'some but not all', (5) meaning will be *John met no student, or John met all students*, and this reading is undesirable. Since in IRM-LU there is such refinement of 'some' in the lexicon L_2, we could suspect this will be also the prediction here.

Another problem of negation is that there seem to be implicatures with no lexical source at sight. For example, the negation in (8).

(8) John didn't meet every student.

$\rightsquigarrow$ John met at least one student.

The lexical account does not provide explanation to this positive implicature.

We will now see whether IRM-LU handles these problems. Thus, we examine how adding the lexical uncertainty affects the pragmatic reasoning of IRM.

5.2.1 Negative 'some'/'all'

Consider for example (8). Simplifying the message in the example to 'not all', I assume the the alternatives here are {'not some', 'not all'} (positive alternatives are omitted to avoid the symmetry problem). The set of possible lexica I use here is similar to all examples involving 'some'/'all', $\Lambda = \{L_1, L_2, L_3\}$, where L_1 is the literal lexicon, in L_2 'some' is refined to 'some but not all', and in L_3 'some' is refined to 'all'. The relevant states induced by the alternatives are $\{\neg\exists, \exists \wedge \neg\forall, \forall\}$. Before computing the IRM-LU predictions, first we compute the predictions of the basic probabilistic IRM of Fox and Katzir (2021). We get the likelihoods in Table 7. In this version of IRM, 'not some' identifies $\neg\exists$, and 'not all' identifies $\exists \wedge \neg\forall$ in a single iteration. This is exactly the desired interpretations for the assessment in (5) and the implicature in (8) respectively.

$p(m\|t)$	'not some'	'not all'
$\neg\exists$	$\frac{1}{2}$	$\frac{1}{2}$
$\exists \wedge \neg\forall$	0	1
$\forall$	0	0

Table 7: Negative 'some'/'all', probabilistic IRM

identified message-state pairs in iteration I
cells eliminated after iteration I

Now we check the IRM-LU results. I skip some intermediate calculations, but for example one can verify that $p(\cdot|\neg\exists, L) = \frac{1}{2}$ for any L since the negation of both 'some' and 'all' satisfies $\neg\exists$. We get the following likelihoods for Λ:

$p(m\|t, L_1)$	'not some'	'not all'
$\neg\exists$	$\frac{1}{2}$	$\frac{1}{2}$
$\exists \wedge \neg\forall$	0	1
$\forall$	0	0
$p(m\|t, L_2)$	'not some'	'not all'
$\neg\exists$	$\frac{1}{2}$	$\frac{1}{2}$
$\exists \wedge \neg\forall$	0	1
$\forall$	1	0
$p(m\|t, L_3)$	'not some'	'not all'
$\neg\exists$	$\frac{1}{2}$	$\frac{1}{2}$
$\exists \wedge \neg\forall$	$\frac{1}{2}$	$\frac{1}{2}$
$\forall$	0	0

To compute the marginalization $p(m|t)$ we note a convenient fact: each row of state t sums to one or it is all zeros. if it is all zeros, then the listener assumes that the speaker will not use this lexicon, thus $p(L|t) = 0$. Else, we average all non-zero rows, by our flat likelihoods assumption in (15). Overall we get:

$p(m\|t)$	'not some'	'not all'
$\neg\exists$	$\frac{1}{2}$	$\frac{1}{2}$
$\exists \wedge \neg\forall$	$\frac{1}{6}$	$\frac{5}{6}$
$\forall$	1	0

Table 8: Negative 'some'/'all', IRM-LU

identified message-state pairs in iteration I
cells eliminated after iteration I

Now, IRM-LU identifies 'not all' with $\exists \wedge \neg\forall$, which is compatible with (8), but it also wrongly identifies 'not some' with $\forall$. This is a serious problem for IRM-LU, which is similar to the undesired divergence under negation of the lexical account.

5.2.2 Negative Conjunction

Another example of negation, similar to (8), is conjunction embedded under negation, as in (17). In a similar manner to the 'some'/'all' example, it has the implicature that one of the conjuncts is true. To the lexical account, the lexical value of "or" contains disjunction and the exclusive-or strengthening, whether in the same lexical value or using lexical ambiguity. This is the same challenge to the lexical account, since the positive implicature in (8) does not seem to have a lexical source.

(17) John did not eat an apple and a banana.

$\rightsquigarrow$ John ate an apple or a banana.

Do derive this example, I notate *John ate an apple* as A, *John ate a banana* as B, and the message is simplified to to 'not (A and B)'. The alternatives in this case are {'not A', 'not B', 'not (A or B)', 'not (A and B)'}. Similarly to CRD, the lexical uncertainty here is resulted from lexical refinements of 'or'. In particular we have the literal lexicon where [[or]] is a disjunction, the lexicon where [[or]] is exlusive or, and the lexicon where [[or]] is identical to 'and'.

Before the results of IRM-LU, it is important to note that unlike 'some'/'all' examples, the basic IBR model of Franke (2011) struggles with deriving the 'exclusive or' implicature of simple disjunctions, even though IBR does derive CRD readings. This is because once the disjunctive state $A \vee B$ is included, 'or' will always identify it. To overcome this issue, I follow Frischoff (2025), and in addition to the induced states $\{\neg A \wedge \neg B,\ A \wedge \neg B,\ B \wedge \neg A\}$ I also include the special exclusive or state $(A \wedge \neg B) \vee (B \wedge \neg A)$. This is not a trivial state to add; it is however a valid state, since it serves as a new equivalence class under the equivalence relation of all-message agreement, as defined in Fox and Katzir (2021).

All in all, after including this special state, and the alternatives above, we get the following results for conjuntion under negation in Table 9. We see that 'not A' identifies $B \wedge \neg A$ and 'not B' identifies $A \wedge \neg B$ as expected. Moreover, 'not (A or B)' identifies the empty state $\neg A \wedge \neg B$ (John ate none), and using the special included state, 'not (A and B)' identifies $(A \wedge \neg B) \vee (B \wedge \neg A)$, deriving the implicature in (17). Unlike the 'some'/'or' example, IRM-LU does derive the SI in conjunction under negation. However, it is only due to the exclusive or state introduced in Frischoff (2025). This state is purely pragmatic and has no lexical source, so the critic about lexical uncertainty remains.

$p(m\|t)$	'not A'	'not B'	'not (A or B)'	'not (A and B)'
$\neg A \wedge \neg B$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$A \wedge \neg B$	0	$\frac{4}{9}$	$\frac{1}{9}$	$\frac{4}{9}$
$B \wedge \neg A$	$\frac{4}{9}$	0	$\frac{1}{9}$	$\frac{4}{9}$
$(A \wedge \neg B) \vee (B \wedge \neg A)$	0	0	$\frac{1}{6}$	$\frac{5}{6}$

Table 9: Conjuction under negation, IRM-LU

cells eliminated after iteration I
identified message-state pairs in iteration I

5.2.3 Negative Free-Choice

Similarly to the suggested IRM in Frischoff (2025), I show that IRM-LU can derive the Negative Free-Choice implicature (Fox, 2007) for two conjuncts.

(18) John is not required to eat an apple and a banana

$\rightsquigarrow$ John is not required to eat an apple

$\rightsquigarrow$ John is not required to eat a banana

In negative free choice, we assume the set of alternatives is not closed to disjunction, so the alternatives here are {'not A', 'not B', 'not (A and B)'}. Hence, the exclusive or state of the last example is not induced. Using the same lexica as before, we get the results in Table 10.

$p(m\|t)$	'not A'	'not B'	'not (A and B)'
$\neg A \wedge \neg B$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$A \wedge \neg B$	0	$\frac{1}{2}$	$\frac{1}{2}$
$B \wedge \neg A$	$\frac{1}{2}$	0	$\frac{1}{2}$

Table 10: Negative Free-Choice, IRM-LU

cells eliminated after iteration I
identified message-state pairs in iteration I
identified message-state pairs in iteration II

Here in the first iteration we have the expected identifications of 'not A' for $B \wedge \neg A$ and 'not B' for $A \wedge \neg B$. After both are eliminated in the first iteration, we are only left with 'not (A and B)' identifying the empty state $\neg A \wedge \neg B$, as required.

5.3 Two Upper Bound Triggering Expressions

I will now introduce another challenge for the lexical ambiguity approach, adapted from Sauerland (2017). Consider the following example:

(19) John met Alice or some smart student.

⇝ John did not meet all students.

⇝ John did not meet both Alice and a smart student.

For the assessment in (19) we have two implicatures. The first comes from the interpretation of 'some' as 'some but not all', and the second from the interpretation of 'or' as 'not both'. To the lexical ambiguity account, both 'some' and 'or' can have two lexical values in the lexicon, one for the literal meaning and one for the upper bound. However, when we combine two expressions that can have upper bound in a single utterance, we might get some undesirable predictions. In (19), if for instance the meaning of 'or' is exclusive or and the meaning of 'some' is some-but-not-all, we get that the sentence is true if John met both Alice and all smart students, which is not the desired meaning.

Another example is two upper bounds of 'or':

(20) John met Mary or Sue or Bill

⇝ John met exactly one person.

Also here, assuming lexical ambiguity, we get wrong predictions. For example if both 'or's are interpreted as exclusive or, we get that the sentecne is true if John met one person or that John met all three. This is of course not the intended meaning of (20).

This is a major challenge to the lexical ambiguity account, and therefore might be a challenge for the suggested IRM-LU model. We would like to examine the model ability to express such messages with more than one possible upper bound.

First, let us note that we had two possible options to define lexical uncertainty, as described in section 4.4. The setting I chose assumes multiple lexica, while each lexicon is consistent. That means, in sentences with two possible upper bounds, the upper bound of each operator is always the same. For example in (20), the model does not permit the meaning where the first 'or' is interpreted as disjunction and the second 'or' as exclusive.

Second, as Franke (2011) shows, the IBR model that Fox and Katzir (2021) use has problems with deriving exclusive or implicature in general. To isolate the role of lexical uncertainty, We must choose a two-upper bound example where the probabilistic IRM of Fox and Katzir (2021) works well. Such example could be free choice disjunction. Consider, for instance, the two upper bounds in (21).

(21) John is allowed to eat an apple or some of the bananas.

⇝ John is allowed to eat an apple.

⇝ John is not allowed to eat all the bananas.

Let us show that the basic probabilistic IRM derives this example. The simplified message here is 'A or some B'. The alternatives, excluding conjunctions, are {'A', 'some B', 'all B', 'A or some B', 'A or all B'} the induces states are a product of the state of A and the possible scale of B. Overall, we get the following probabilities in IRM:

$p(m\|t)$	'A'	'some B'	'all B'	'A or some B'	'A or all B'
$A \wedge \neg\exists B$	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{3}$
$A \wedge (\exists \wedge \neg\forall) B$	$\frac{1}{4}$	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{4}$
$A \wedge \forall B$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
$(\neg A) \wedge \neg\exists B$	0	0	0	0	0
$(\neg A) \wedge (\exists \wedge \neg\forall) B$	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0
$(\neg A) \wedge \forall B$	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Table 11: Two free-choice upper bounds, probabilistic IRM

cells eliminated after iteration I
cells eliminated after iteration II
cells eliminated after iteration III
identified message-state pairs in iteration I
identified message-state pairs in iteration II
identified message-state pairs in iteration III

We can see that in the first iteration, 'some' identifies $\neg A \wedge (\exists \wedge \neg\forall) B$, 'A' identifies $A \wedge \neg\exists B$, and 'all B' identifies $\neg A \wedge \forall B$. After the first elimination, in the second iteration 'A or some B' identifies $A \wedge (\exists \wedge \neg\forall) B$, which is the intended meaning of (21).

Does the addition of lexical uncertainty hurt the probabilistic IRM prediction for (21)? First we need to understand what is the lexical uncertainty for IRM-LU here. Since there are known three refinements for 'some' and three refinements for 'or', using all possible refinements over the literal lexicon will result in the product of nine lexica in Λ. Marginalizing over all lexica, we obtain the following probabilities (I use numerical values here for convenience):

$p(m\|t)$	'A'	'some B'	'all B'	'A or some B'	'A or all B'
$A \wedge \neg\exists B$	0.56	0	0	0.22	0.22
$A \wedge (\exists \wedge \neg\forall) B$	0.39	0.20	0	0.20	0.20
$A \wedge \forall B$	0.26	0.16	0.26	0.15	0.15
$(\neg A) \wedge \neg\exists B$	0	0	0	0	0
$(\neg A) \wedge (\exists \wedge \neg\forall) B$	0	0.67	0	0.33	0
$(\neg A) \wedge \forall B$	0	0.22	0.44	0.11	0.22

Table 12: Two free-choice upper bounds, IRM-LU

cells eliminated after iteration I
cells eliminated after iteration II
cells eliminated after iteration III
identified message-state pairs in iteration I
identified message-state pairs in iteration II
identified message-state pairs in iteration III

In IRM-LU we see the exact same pattern of state identification as the probabilistic IRM of Fox and Katzir (2021). In particular, IRM-LU correctly derives (21) that has two upper-bound triggering expressions. Overall, the two-upper-bound problem, that was a major challenge to the lexical ambiguity account, seems to be handled under IRM-LU. Yet, we showed it only for free choice disjunction. As mentioned, all IBR models based on Franke (2011) still struggle with exclusive or, and in particular the two-upper-bound examples in (19) and (20).

5.4 Ellipsis

The last challenge for the lexical ambiguity approach is ellipsis, and in particular the requirement of ellipsis for semantic identity between the pronounced antecedent and the elided part. Consider the example in (22).

(22) John met some of the students and Mary didn't ~~meet some of the students~~.

$\rightsquigarrow$ John met some but not all students.

$\rightsquigarrow$ Mary did not meet any student.

Here we have two interpretations for 'some'. The first, in the UE context, is the upper bound 'some but not all', thus John did not meet all students. However in the DE context of the elided part, 'some' does not get the upper bound interpretation, and the intuitive reading is that Mary did not meet any student.

This is a problem for the lexical ambiguity account. If the semantic value of 'some' is interpreted literally both times, we get that (22) is true if John met all students. If the semantic value of 'some' gets the upper-bound interpretation both times, we get that (22) is true if Mary met all students. Lastly, if 'some' gets different semantic values in the UE antecedent and the elided DE part, we violate the requirement for semantic identity in ellipsis.

Given this data, we would expect the lexical uncertainty of IRM-LU to tackle the same challenge. Nevertheless, I will now show that IRM-LU does derive the desired reading for (22), while keeping the semantic identity constraint.

Recall that due to our requirement in the definition of IRM-LU to lexical consistency within the same lexicon, the semantic identity of ellipsis is not a problem, and every interpreted message has self-consistency. In fact this is another argument against the setting that was suggested in section 4.4.

To derive (22), the simplified message including the elided part will be 'some J and not (some M)' where 'J' notates events of John meeting the students, and 'M' the same for Mary. To derive the alternative messages, we have two structural constraints: the first is the semantic identity of ellipsis (so 'some' in the first part implies 'some' in the elided part, and same for 'all'), and the second is not removing negation, as in section 5.2. Overall, we get the following alternatives: {'some J', 'all J', 'not (some M)', 'not (all M)', 'some J and not (some M)', 'all J and not (all M)', 'some J or not (some M)', 'all J or not (all M)'}. The relevant states are a product of the 'some' scales of 'J' and 'M'. The

lexica we use for the lexical uncertainty are the same three lexica that are used for 'some'. After the marginalization, finally we get the following results:

$p(m\|t)$	'some J'	'all J'	'not (some M)'	'not (all M)'
$\neg\exists J \wedge \neg\exists M$	0	0	$\frac{1}{4}$	$\frac{1}{4}$
$\neg\exists J \wedge (\exists \wedge \neg\forall) M$	0	0	$\frac{1}{12}$	$\frac{5}{12}$
$\neg\exists J \wedge \forall M$	0	0	$\frac{1}{2}$	0
$(\exists \wedge \neg\forall) J \wedge \neg\exists M$	$\frac{1}{9}$	0	$\frac{7}{36}$	$\frac{7}{36}$
$(\exists \wedge \neg\forall) J \wedge (\exists \wedge \neg\forall) M$	$\frac{1}{6}$	0	$\frac{1}{12}$	$\frac{1}{4}$
$(\exists \wedge \neg\forall) J \wedge \forall M$	$\frac{3}{8}$	0	$\frac{1}{8}$	0
$\forall J \wedge \neg\exists M$	$\frac{1}{12}$	$\frac{5}{36}$	$\frac{5}{36}$	$\frac{5}{36}$
$\forall J \wedge (\exists \wedge \neg\forall) M$	$\frac{7}{72}$	$\frac{13}{72}$	$\frac{1}{24}$	$\frac{13}{72}$
$\forall J \wedge \forall M$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{12}$	0

$p(m\|t)$	'some J and not (some M)'	'some J or not (some M)'	'all J and not (all M)'	'all J or not (all M)'
$\neg\exists J \wedge \neg\exists M$	0	$\frac{1}{4}$	0	$\frac{1}{4}$
$\neg\exists J \wedge (\exists \wedge \neg\forall) M$	0	$\frac{1}{12}$	0	$\frac{5}{12}$
$\neg\exists J \wedge \forall M$	0	$\frac{1}{2}$	0	0
$(\exists \wedge \neg\forall) J \wedge \neg\exists M$	$\frac{1}{9}$	$\frac{7}{36}$	0	$\frac{7}{36}$
$(\exists \wedge \neg\forall) J \wedge (\exists \wedge \neg\forall) M$	0	$\frac{1}{4}$	0	$\frac{1}{4}$
$(\exists \wedge \neg\forall) J \wedge \forall M$	$\frac{1}{8}$	$\frac{3}{8}$	0	0
$\forall J \wedge \neg\exists M$	$\frac{1}{12}$	$\frac{5}{36}$	$\frac{5}{36}$	$\frac{5}{36}$
$\forall J \wedge (\exists \wedge \neg\forall) M$	$\frac{1}{24}$	$\frac{7}{72}$	$\frac{13}{72}$	$\frac{13}{72}$
$\forall J \wedge \forall M$	0	$\frac{1}{4}$	0	$\frac{1}{4}$

Table 13: Ellipsis in (22) with IRM-LU

cells eliminated after iteration I
cells eliminated after iteration II
identified message-state pairs in iteration I
identified message-state pairs in iteration II

While the probabilities and the state identifications are quite messy, we see that the message 'some J and not (some M)' identifies the state $(\exists \wedge \neg\forall) J \wedge \neg\exists M$ in the second iteration. This is exactly the desired meaning of (22): John met some but not all students, and Mary did not meet any student. Thus, we showed that IRM-LU derives ellipsis while keeping semantic identity.

6 Discussion

I opened this work by introducing the simplest account for scalar implicatures, which is the lexical account. The lexical account however struggles to explain simple phenomena such as implicature suspension or no upper bounds in DE contexts. We shifted to a lexical ambiguity approach, that holds different semantic values for literal meanings and upper bounds. This approach too face many challenges, including divergence under negation, implicatures with no lexical source at sight, two-upper bound triggering expressions and semantic identity in ellipsis. These challenges are arguments in favor of advanced theories for SI, such as the neo-Gricean approach of pragmatic reasoning (Sauerland, 2004) or the grammatical approach of local exhaustification (Chierchia, Fox, and Spector, 2012).

The goal of this work was to examine whether methods in the pragmatic IRM literature that suggest use of multiple semantic values, such as Bergen, Levy, and Goodman (2016), face the same challenges of lexical ambiguity. For doing so, I constructed a new model merging IRM ideas from Fox and Katzir (2021) and lexical uncertainty from Bergen, Levy, and Goodman (2016), named IRM-LU.

The results indicate that lexical uncertainty, in the sense of Bergen, Levy, and Goodman (2016) that keeps self lexical consistency within the same lexicon, is quite different from naive lexical ambiguity in its predictions. An IRM with naive lexical ambiguity, such as the one that was suggested in section 4.4, fails to derive CRD, and will probably struggle with ellipsis, or two-upper-bound triggering expressions. On the other hand, the use of lexical uncertainty, where all possible lexical refinements are pragmatically combined to create a single interpretation, does not seem to tackle these problems. The major problem for IRM-LU was unexpected divergence under negation for negative 'some'/'all'. Yet, IRM-LU seems to derive non trivial results, such as CRD with more than two disjuncts, and to face most of the challenges for the lexical ambiguity approach.

Despite the success of IRM-LU predictions in solving lexical ambiguity challenges, there are several arguments against this model as a theory of scalar implicatures.

First, lexical ambiguity is usually assumed for words that have true semantic ambiguity, such as "bank". The same should apply for lexical uncertainty. The fact that each scalar item has multiple meanings, and that it is a cross-lingual phenomenon, does not seem to be a coincidence. Storing many lexical refinements for each scalar item creates unnecessary cumbersome semantics, which hurts the explanatory adequacy of the theory.

Second, the theory of Bergen, Levy, and Goodman (2016), and my following suggested model, seems to break the separation between semantics and pragmatics. The theory of lexical uncertainty assumes that the listener infers the lexical content only in the process of pragmatic inference. While Bergen, Levy, and Goodman (2016) advocate this view, this assumption might be controversial. In fact, most of the results of IRM-LU are resulted due to pure pragmatic

reasoning, and not due to lexical uncertainty. The evidence comes from the fact that the original probabilistic IRM of Fox and Katzir (2021) derives the same predictions for many of the cases that were examined in this work.

After presenting the probabilistic IRM, Fox and Katzir (2021) provide arguments against it. The first argument is based on the inability of probabilistic IRM to derive CRD for more than two disjuncts. This argument was later refuted by Frischoff (2025), and also in this work I showed that IRM-LU can derive CRD for three disjuncts, however without proof for the general case. The second argument of Fox and Katzir (2021) is based on prior sensitivity of IRM pragmatics. This argument is still valid, as further demonstrated by Frischoff (2025), and requires future research.

One of the main arguments in favor of the grammatical approach to SI over the neo-Gricean approach that is implemented in the IRM literature, is the existence of local readings of embedded implicatures (Chemla and Spector, 2011). A following work to Bergen, Levy, and Goodman (2016) by Potts et al. (2016) adapted the compositional lexical uncertainty model to derive these local readings, suggesting a pragmatic view of these implicatures. An adaptation of the ideas of Potts et al. (2016) and proper examination of IRM-LU for embedded implicatures is a possible future work.

7 Conclusion

I presented IRM with lexical uncertainty, a pragmatic reasoning model to derive scalar implicatures merging the suggestions of Fox and Katzir (2021) and Bergen, Levy, and Goodman (2016). I showed that while a naive lexical ambiguity account for SI faces multiple challenges, the predictions of IRM-LU usually coincide with the natural interpretations of SI in these cases. In particular, IRM-LU well-predicts possible implicature suspension, two upper-bound triggering expressions and ellipsis, but struggles with 'some'/'all' negation as it predicts undesired divergence. In addition, IRM-LU predicts CRD for three disjuncts, in contrast to the original probabilistic IRM of Fox and Katzir (2021). Lastly, I discussed some arguments against IRM-LU and lexical uncertainty.

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